

# Kinematics of the Passive Split Spherical Caster

## 1 Geometry and Coordinate Definitions

Let the caster-fixed coordinate frame be  $\{C\}$  with basis  $\{\hat{x}_c, \hat{y}_c, \hat{z}_c\}$ . The common rod is aligned with

$$\hat{y}_c = [0 \quad 1 \quad 0]^T. \quad (1)$$

The spherical shell has radius  $R$ . Let  $\alpha$  denote the orientation of the whole spherical-caster assembly about the common rod.

At  $\alpha = 0$ , the hemisphere's own spin axis is vertical, so

$$\mathbf{a}_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (2)$$

After the assembly rotates about  $\hat{y}_c$  by  $\alpha$ ,

$$\mathbf{a}(\alpha) = R_y(\alpha)\mathbf{a}_0, \quad (3)$$

where

$$R_y(\alpha) = \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix}. \quad (4)$$

Therefore,

$$\boxed{\mathbf{a}(\alpha) = \begin{bmatrix} \sin \alpha \\ 0 \\ \cos \alpha \end{bmatrix}}. \quad (5)$$

## 2 Ideal Contact Geometry and Effective Rotational Radius

For the ideal kinematic model, assume the line from the caster center  $O$  to the ground contact point  $C$  is normal to the ground. Then

$$\mathbf{r}_{OC} = \begin{bmatrix} 0 \\ 0 \\ -R \end{bmatrix} = -R\hat{z}_c. \quad (6)$$

The effective rotational radius of the hemisphere about its own spin axis is the perpendicular distance from the spin axis to the contact point:

$$\rho(\alpha) = \|\mathbf{a}(\alpha) \times \mathbf{r}_{OC}\| = R|\sin \alpha|. \quad (7)$$

For a local operating range with  $\sin \alpha > 0$ , this becomes

$$\boxed{\rho(\alpha) = R \sin \alpha}. \quad (8)$$

Hence,

$$\rho(0) = 0, \quad \rho\left(\frac{\pi}{2}\right) = R. \quad (9)$$

The configuration  $\alpha = 0$  is therefore a kinematic singularity for motion generated by hemisphere spin.

### 3 Angular Velocity and No-Slip Kinematics

Let

- $\dot{\alpha}$  be the common-rod rotation rate about  $\hat{\mathbf{y}}_c$ ,
- $\dot{\beta}$  be the hemisphere spin rate about  $\mathbf{a}(\alpha)$ .

Choose the sign convention for  $\beta$  so that positive  $\dot{\beta}$  generates positive  $y_c$  translation. The hemisphere angular velocity is then

$$\boxed{\boldsymbol{\omega} = \dot{\alpha} \hat{\mathbf{y}}_c - \dot{\beta} \mathbf{a}(\alpha)}. \quad (10)$$

For ideal no-slip contact,

$$\mathbf{v}_{\text{contact}} = \mathbf{v}_C + \boldsymbol{\omega} \times \mathbf{r}_{OC} = \mathbf{0}, \quad (11)$$

so the caster-center velocity is

$$\mathbf{v}_C = -\boldsymbol{\omega} \times \mathbf{r}_{OC}. \quad (12)$$

Substituting the expressions above gives

$$\mathbf{v}_C = -\left[\dot{\alpha} \hat{\mathbf{y}}_c - \dot{\beta} (\sin \alpha \hat{\mathbf{x}}_c + \cos \alpha \hat{\mathbf{z}}_c)\right] \times (-R \hat{\mathbf{z}}_c) \quad (13)$$

$$= R \dot{\alpha} \hat{\mathbf{x}}_c + R \dot{\beta} \sin \alpha \hat{\mathbf{y}}_c. \quad (14)$$

Therefore,

$$\boxed{v_x = R \dot{\alpha}}, \quad \boxed{v_y = R \sin \alpha \dot{\beta} = \rho(\alpha) \dot{\beta}}. \quad (15)$$

### 4 Caster Jacobian and Singularity

Define the caster state

$$\mathbf{q}_c = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}. \quad (16)$$

The caster-frame velocity satisfies

$$\begin{bmatrix} v_x \\ v_y \end{bmatrix} = J_c(\alpha) \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \end{bmatrix}, \quad (17)$$

with

$$\boxed{J_c(\alpha) = \begin{bmatrix} R & 0 \\ 0 & R \sin \alpha \end{bmatrix}}. \quad (18)$$

Away from the singularity,

$$J_c^{-1}(\alpha) = \begin{bmatrix} \frac{1}{R} & 0 \\ 0 & \frac{1}{R \sin \alpha} \end{bmatrix}, \quad (19)$$

so

$$\boxed{\dot{\alpha} = \frac{v_x}{R}}, \quad \boxed{\dot{\beta} = \frac{v_y}{R \sin \alpha}}. \quad (20)$$

Also,

$$\det J_c = R^2 \sin \alpha. \quad (21)$$

Thus, for  $\sin \alpha \neq 0$ ,  $\text{rank}(J_c) = 2$ , whereas at  $\alpha = 0$  (and more generally at  $\sin \alpha = 0$ ),

$$\det J_c = 0, \quad \text{rank}(J_c) = 1. \quad (22)$$

## 5 World-Frame Kinematics

Let  $\psi$  be the yaw of the caster frame relative to the world frame. Then

$$\mathbf{v}_W = R_z(\psi) \mathbf{v}_C, \quad (23)$$

where

$$R_z(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix}. \quad (24)$$

Therefore,

$$\boxed{\begin{bmatrix} \dot{X} \\ \dot{Y} \end{bmatrix} = R \begin{bmatrix} \cos \psi & -\sin \psi \sin \alpha \\ \sin \psi & \cos \psi \sin \alpha \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \end{bmatrix}}. \quad (25)$$

Equivalently,

$$\dot{X} = R \cos \psi \dot{\alpha} - R \sin \psi \sin \alpha \dot{\beta}, \quad (26)$$

$$\dot{Y} = R \sin \psi \dot{\alpha} + R \cos \psi \sin \alpha \dot{\beta}. \quad (27)$$

## 6 Integrated Motion for Constant Rates

Consider the caster frame ( $\psi = 0$ ) with constant nonzero  $\dot{\alpha}$  and constant  $\dot{\beta}$ . Then

$$\alpha(t) = \alpha_0 + \dot{\alpha}t, \quad (28)$$

with

$$\dot{x} = R\dot{\alpha}, \quad \dot{y} = R\dot{\beta} \sin(\alpha_0 + \dot{\alpha}t). \quad (29)$$

Integrating,

$$\boxed{x(t) = x_0 + R\dot{\alpha}t}, \quad (30)$$

and, for  $\dot{\alpha} \neq 0$ ,

$$\boxed{y(t) = y_0 + \frac{R\dot{\beta}}{\dot{\alpha}} [\cos \alpha_0 - \cos(\alpha_0 + \dot{\alpha}t)]}. \quad (31)$$

Since

$$t = \frac{x - x_0}{R\dot{\alpha}}, \quad (32)$$